

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
In [2]: df = pd.read_csv("Titanic-Dataset.csv")
df.head()
```

Out:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
In [3]: df.info()
```

Out:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null   int64
1   Survived    891 non-null   int64
```

```
2  Pclass      891 non-null   int64
3  Name        891 non-null   object
4  Sex         891 non-null   object
5  Age         714 non-null   float64
6  SibSp       891 non-null   int64
7  Parch       891 non-null   int64
8  Ticket      891 non-null   object
9  Fare        891 non-null   float64
10 Cabin       204 non-null   object
11 Embarked    889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [4]: df.describe()

Out:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [5]: df.isnull().sum()

Out:

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

In [6]: df['Age'] = df['Age'].fillna(df['Age'].median())

```

In [7]: df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

In [8]: df.drop('Cabin', axis=1, inplace=True)

In [9]: le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])

In [10]: X = df[['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']]
y = df['Survived']

In [11]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

In [12]: model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

```

Out:

▼ LogisticRegression ⓘ ?

► Parameters

```

In [13]: y_pred = model.predict(X_test)

In [14]: accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

Out:
Accuracy: 0.8100558659217877

In [15]: print(classification_report(y_test, y_pred))

```

Out:

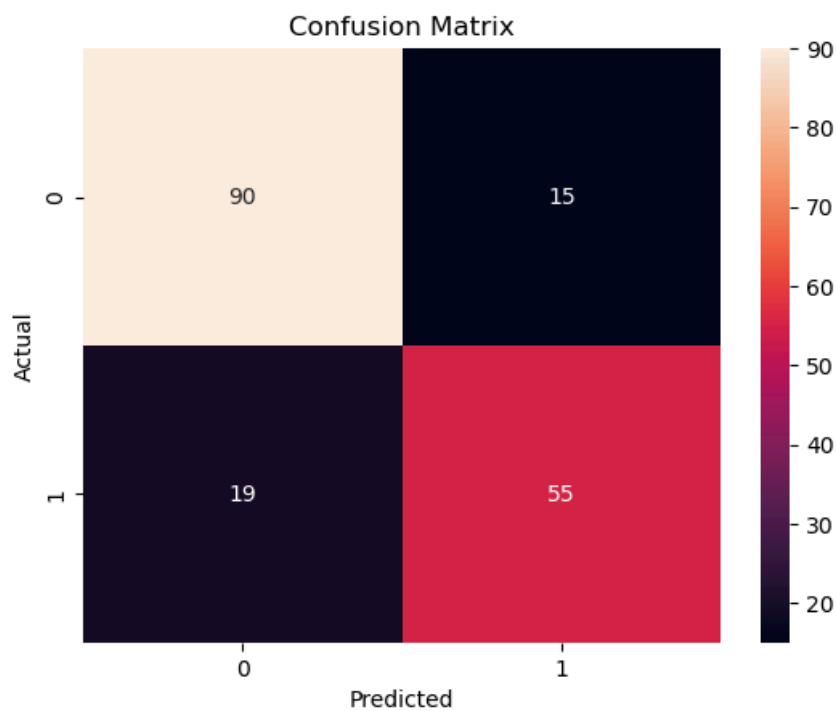
	precision	recall	f1-score	support
0	0.83	0.86	0.84	105

	1	0.79	0.74	0.76	74
accuracy				0.81	179
macro avg	0.81	0.80	0.80		179
weighted avg	0.81	0.81	0.81		179

```
In [16]: cm = confusion_matrix(y_test, y_pred)
```

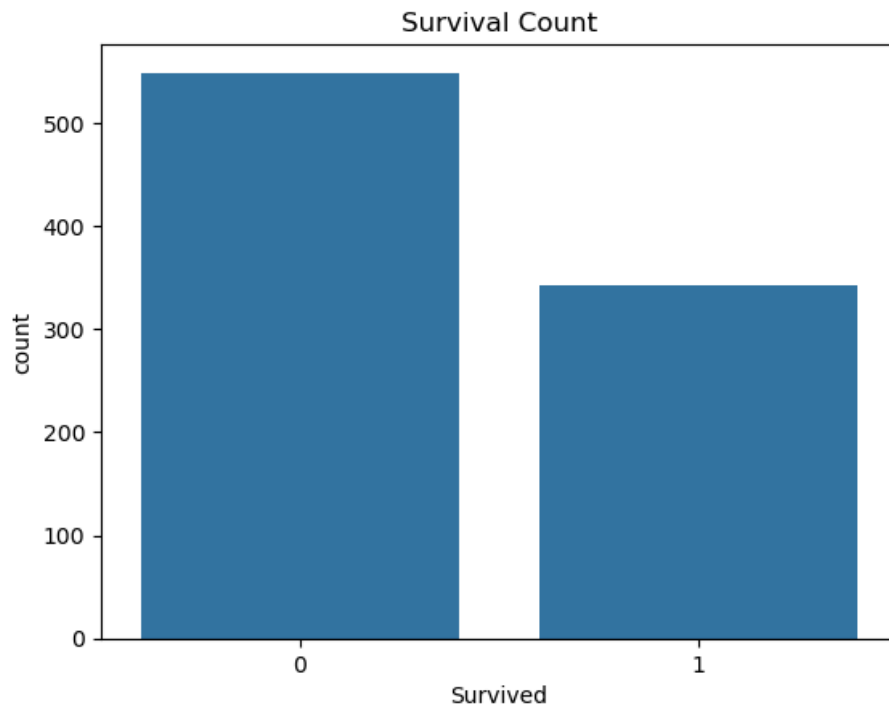
```
sns.heatmap(cm, annot=True, fmt='d')  
plt.xlabel("Predicted")  
plt.ylabel("Actual")  
plt.title("Confusion Matrix")  
plt.show()
```

Out:



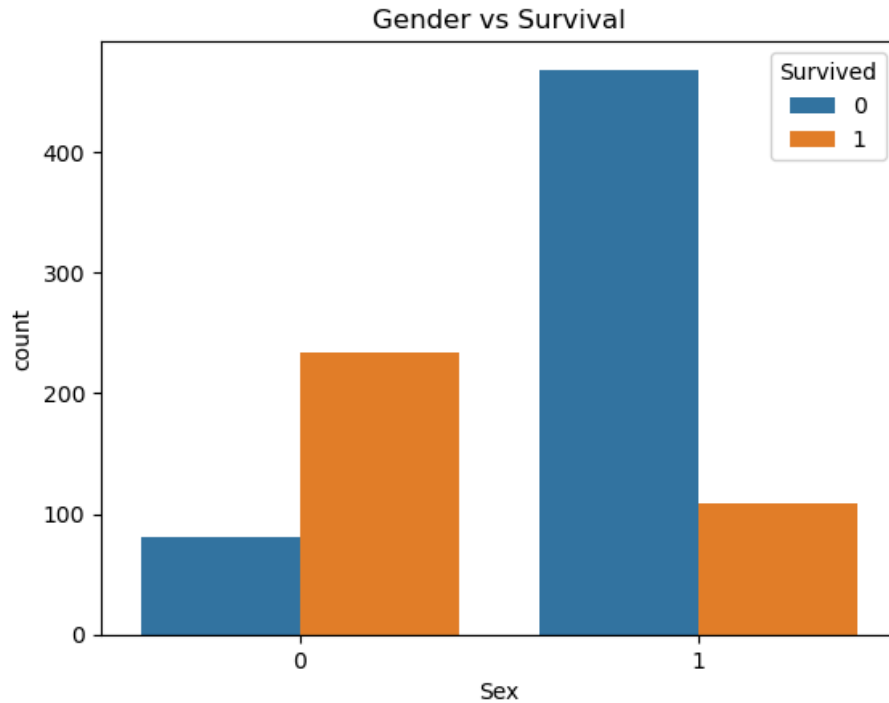
```
In [17]: sns.countplot(x='Survived', data=df)  
plt.title("Survival Count")  
plt.show()
```

Out:



```
In [18]: sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Gender vs Survival")
plt.show()
```

Out:



```
In [19]: sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Passenger Class vs Survival")
plt.show()
```

Out :

